CO2 AT1

Analytical problem solving

Name: A.Midhuna

Register No: 192521302

Course Name: Cryptography and network Security

1. Soln:

RSA Key Generation and Security Analysis

$p = 17, q = 19$

public exponent = 5

$m = 7$

calculate n:

$n = p \times q = 17 \times 19$

$n = 323$

calculate Euler's Totient Function

$\phi(n) = (p-1)(q-1)$

$\phi(n) = 16 \times 18$

$\phi(n) = 288$

calculate private key d

$(e \times d) \bmod \phi(n) = 1$

$(5 \times d) \bmod 288 = 1$

$d = 173$ $\quad 5 \times 173 = 865$

$865 \bmod 288 = 1$

$d = 173$

public key $(e, n) = (5, 323)$

private key $(d, n) = (173, 323)$

Step 4: Encrypt the message

RSA encryption formula

$$c = m^e \bmod n$$

$$c = 7^5 \bmod 323$$

$$7^5 = 16807$$

$$16807 \bmod 323 = 280$$

ciphertext : $c = 280$

Step 5 : Decrypt the ciphertext :

RSA decryption formula :

$$m = c^d \bmod n$$

$$m = 280^{173} \bmod 323$$

$$m = 7$$

Verification : The decrypted message matches the original message $m=7$. encryption and decryption are correct.

Step 6: Security Analysis of small prime numbers

* Easy factorization : small primes make n small, so attackers can easily find p and q.

* private key recovery : If the prime factors are known an attacker calculate $\varphi(n)$ and determine the private key.

* Brute force attacks : small RSA keys more vulnerable guessing computational attacks.

* Data confidentiality: An attacker may recover encrypted information key compromised.

* Increased security: Using very large prime numbers increases difficulty factoring n.

* Key size: Real world RSA commonly used a modulus least 2048 bits.

* Educational purpose: The small values used in suitable learning RSA calculations but not for protecting real data.

conclusion:

RSA uses a public key encryption private key decryption. The ciphertext 280 and decryption successfully recover the original message 7. However using small prime numbers makes RSA insecure because attacks factor modulo and recover private key.

2. ElGamal Encryption Analysis:

* prime number $p = 23$
* Generator $g = 5$
* private key: $x = 7$
* Random number $K = 3$
* message $M = 10$

Step 1: compute the public key

$$y = g^x \bmod p$$

$$y = 5^7 \bmod 23$$

$$5^7 \bmod 23 = 17$$

$$(p, g, y) = (23, 5, 17)$$

private key $x = 7$

Step 2 : Encrypt the message

ElGamal encryption generates two ciphertext values.

calculate c_1 :

$$c_1 = g^k \bmod p$$

$$c_1 = 5^3 \bmod 23$$

$$c_1 = 10$$

calculate c_2 :

$$c_2 = m \times y^k \bmod p$$

$$c_2 = 10 \times 17^3 \bmod 23$$

$$17^3 \bmod 23 = 14$$

$$c_2 = 10 \times 14 \bmod 23$$

$$c_2 = 2$$

ciphertext pair

$$(c_1, c_2) = (10, 2)$$

Step 3 : Decrypt ciphertext

$$m = c_2 \times (c_1^x)^{-1} \bmod p$$

$$c_1^x = 10^7 \bmod 23 = 14$$

The modular inverse 14 modulo 2^3 is 5 because.

$14 \times 5 \bmod 23 = 1$

$m = 2 \times 5 \bmod 23$

$m = 10$

Verification:

The decrypted message 10 which matches the original message. Hence encryption and decryption correct.

Step 4: Security Analysis of Randomness

* Improve security: Random K makes encryption harder predict.
* Different ciphertext: The same message can produce different ciphertext different value K used.
* prevents pattern detection: Randomness makes harder attacks identify repeated messages by comparing ciphertexts.
* must remain secret: If K is exposed attack may recover message or private key.
* Never reuse K: Reusing the same K reveal relationship message and weaken security.
* Use Secure randomness: The value K must be generated using cryptoghically secure random generator.
* valid range: If example K = 3 within the

valid range $1 \leq K \leq p-2$

* practical security: Real implementations use much larger groups are carefully designed cryptographic parameters

conclusion:

Elgamal encryption produces the ciphertext pair (10, 2) successfully decrypts it recover original message 10. Randomness K improves security by making encryption less predictable provided it is generated securely and never reused.